

Reliability-Aware Multiscale Design of Functionally Graded Perforated Plate Metamaterials Under Static, Buckling, and Modal Constraints

Research Proposal and Paper Draft

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Abstract

Perforated plate metamaterials offer a continuous and manufacturable route to tunable stiffness, strength, and vibration characteristics, but existing design methodologies treat nominal geometry as deterministic and largely confine structural evaluations to static loading alone. This study proposes and executes a reliability-aware, multiscale computational framework for the design of functionally graded perforated plate metamaterials subject to simultaneous static, eigenvalue buckling, and free-vibration (modal) constraints. A parametric family of elliptically perforated rectangular coupons is characterized by five geometric descriptors — void fraction ϕ , hole aspect ratio $\eta = a/b$, orientation θ , pitch ratio p_x/p_y , and minimum ligament ratio λ — and evaluated using the Autodesk Fusion 360 finite element solver across a Latin-hypercube design-of-experiments campaign. Cell-scale effective properties are extracted through a computational homogenization procedure employing kinematically uniform boundary conditions, yielding the effective stiffness tensor $\mathbb{C}^{\text{eff}}$ and an anisotropy index that bridges micro-scale geometry to macro-scale structural response. A Gaussian process surrogate model is trained on the simulation ensemble, and a Non-dominated Sorting Genetic Algorithm (NSGA-II) identifies Pareto-optimal designs with respect to mass-normalized specific stiffness, buckling multiplier, and first natural frequency. Functional grading is subsequently introduced through a low-dimensional porosity field, and the nominal optimum is contrasted with a robust optimum obtained by propagating truncated-normal geometric uncertainties through the surrogate. Variance-based Sobol sensitivity indices identify the dominant uncertain parameters. The framework demonstrates that a modestly graded perforation pattern outperforms uniform designs at equal mass within the combined static–buckling–modal trade space, and that robust optimization sacrifices limited nominal performance while substantially reducing failure probability and output variance — quantitative findings that remain informative regardless of their magnitude.^[1]

Keywords: Perforated plate metamaterials; Functionally graded structures; Finite element analysis; Homogenization; Multiobjective optimization; Uncertainty quantification; Reliability-based design; Buckling; Modal analysis

1. Introduction

1.1 Motivation and Background

Lightweight structural architectures that achieve high specific stiffness and damage resistance are essential across aerospace, civil, and biomedical engineering. Mechanical metamaterials — periodic or quasi-periodic microstructured solids whose effective properties are governed by geometry rather than chemistry — represent a rapidly expanding design paradigm for this purpose. Among the various metamaterial topologies explored in the literature, perforated plate architectures are distinguished by their practical manufacturability, continuous tunability, and structural efficiency. Plate lattices exhibit stiffness and yield strength superior to shell and truss lattices across a broad relative density range, and introducing through-thickness perforations directly into plate faces can improve manufacturability — particularly for additive-manufactured components produced via laser powder bed fusion — without catastrophically degrading mechanical performance.^{[2][3]}

The design of such structures, however, presents non-trivial challenges. First, performance in engineering systems is governed not only by static compliance but also by resistance to elastic instability and by dynamic response characteristics. Buckling failure is a critical concern in low-density plate lattices, where the critical buckling stress scales cubically with relative density, making the structure acutely sensitive to geometric perturbations in the perforation pattern. At the same time, recent lattice reviews note that the published literature is considerably richer in static characterization than in dynamic behavior, leaving the multi-physics trade space poorly mapped. Second, real manufactured structures inevitably deviate from nominal CAD geometry due to process variability. Additive manufacturing — the primary fabrication route for architected metamaterials — introduces dimensional tolerances and surface irregularities that translate directly into uncertainty in mechanical performance. A design optimized at nominal geometry may therefore be fragile in service.^{[3][4][5][1]}

1.2 Research Gaps

Three compounding gaps motivate this work:

1. **Multi-objective characterization gap.** The majority of perforated plate studies are confined to static stiffness or single-mode buckling analysis. A simultaneous, normalized treatment of static compliance, buckling resistance, and free-vibration frequency as competing objectives within a unified surrogate-optimization loop has not been established for elliptically perforated plate geometries.^[^1]

2. **Multiscale structure–property relationship gap.** Direct finite element analysis of full periodic structures is computationally expensive, and practical homogenization-based workflows linking cell-scale geometry to structure-scale performance indices remain underdeveloped for accessible CAD-driven FEM environments. The effective orthotropic stiffness tensor of a perforated cell and its correlation with macro-scale buckling and modal response have not been systematically mapped as a function of the full five-parameter geometry space.^[^1]
3. **Robustness and reliability gap.** Reliability-based design optimization (RBDO) for combined static–buckling–modal performance under geometric manufacturing uncertainty is largely absent from the perforated-plate metamaterial literature. While RBDO under buckling and compliance constraints has been developed for continuum topology optimization, and surrogate-model-based approaches exist for random field load uncertainty, their extension to geometry-driven metamaterial design remains an open problem.^{[4][6][7][8]}

1.3 Objectives and Scope

This study addresses all three gaps through a layered, modular research program organized into two nested scopes:^[^1]

- **Scope A (minimum publishable unit):** Uniform-geometry DOE campaign, surrogate model construction, and Pareto optimization for a five-parameter elliptical perforation family under static, buckling, and modal constraints.
- **Scope B (full contribution):** Functionally graded porosity extension, cell-scale computational homogenization, and robust optimization under truncated-normal geometric uncertainty with Sobol sensitivity decomposition.

The four governing research questions are:

RQ1. Which geometric descriptors most strongly control the trade-off among specific stiffness, buckling resistance, and first natural frequency?

RQ2. Can a low-dimensional graded porosity law outperform the best uniform design at equal mass in the combined static–buckling–modal objective space?

RQ3. How much nominal performance is sacrificed when the optimization shifts from a deterministic to a robust formulation?

RQ4. Which uncertain parameters dominate output variance — hole semi-axes, orientation, ligament width, or base material modulus?

2. Literature Review

2.1 Perforated Plate Metamaterials

Plate lattices occupy a distinct position among metamaterial topologies. Their closed-cell geometry achieves stretch-dominated deformation, conferring higher specific stiffness and yielding strength than open-cell truss or shell counterparts in the moderate-to-high relative density regime. A pivotal 2025 study on metallic perforated plate lattices — fabricated via micro laser powder bed fusion — demonstrated that Rayleigh-quotient-guided hole placement can increase critical buckling stresses over unperforated baselines while preserving post-buckling compressive strength, and that Young's modulus and compressive strength remain superior to shell and truss topologies across relative densities spanning 5.6–37.1%. The same work highlights that in the low-density regime, buckling — not yield — governs load-bearing capacity, underscoring the necessity of buckling-explicit design methods. Contemporaneously, shear post-buckling analysis of perforated plates has revealed that post-buckling bearing capacity increases with plate thickness, that an optimal aperture exists for a given plate dimension, and that geometric reinforcement can increase post-buckling capacity by 7–133%.^{[9][3]}

On the dynamics side, perforated and functionally graded plates exhibit coupled vibration–buckling sensitivity to porosity distribution, graphene reinforcement gradient, and boundary conditions. Symmetric surface-enriched graphene distributions achieve optimal stiffness and natural frequencies, while power-law graded reinforcement outperforms layered distributions in buckling response. These findings motivate the incorporation of a polynomial or zonal porosity grading law as an independent design variable, rather than treating grading as an ad hoc post-processing step.^[^10]

2.2 Functionally Graded Structures

Functionally graded materials (FGMs) are characterized by spatially continuous or stepwise variations in composition or microstructure. In the context of structural mechanics, grading typically serves to reduce stress concentration, delay buckling onset, and shift natural frequencies away from excitation bands. Perforated functionally graded plates have been studied analytically; careful selection of the material variation parameter and grading direction significantly reduces stress and strain concentrations around cutouts. Recent work on graphene-reinforced FG porous plates confirms that the grading profile — not merely the average porosity — critically determines bending, buckling, and vibration response. A 2026 overview of functionally graded engineering structures identifies pore-distribution optimization and multiphysics grading laws as open research directions. The present work contributes a physics-informed, low-dimensional grading law derived directly from FE field outputs, offering a principled alternative to empirical power-law distributions.^{[11][12][^10]}

2.3 Computational Homogenization for Lattice Metamaterials

Computational homogenization extracts macroscopic effective properties from representative volume element (RVE) simulations, enabling separation of cell-scale design from structure-scale analysis. Kinematically uniform boundary conditions prescribed on the RVE yield volume-averaged stress and strain tensors from which the effective stiffness tensor is assembled. Recent work on micromorphic homogenization demonstrates that standard first-order homogenization can be enriched with higher-order deformation modes harvested from experimental observation to capture size-effect behavior in metamaterials. However, for the geometry-driven, elastic metamaterials considered here — where property contrast is achieved through void geometry rather than material heterogeneity — first-order homogenization under kinematically uniform boundary conditions is sufficient and well-validated. A 2025 multiscale lattice design paper notes that a practical homogenization-based operational procedure linking repeated-cell topology to structure-scale predictions is still missing from the accessible design toolchain, which this study directly addresses.^{[13][14][1]}

2.4 Surrogate-Based Optimization

Direct multi-objective optimization over computationally expensive FE simulation landscapes requires surrogate (metamodel) approximations. Gaussian process (Kriging) surrogates are the dominant choice for mechanical design optimization due to their interpolation exactness, built-in uncertainty quantification, and suitability for adaptive enrichment via expected improvement criteria. In the context of structural optimization under uncertainty, polynomial chaos expansion (PCE) has been deployed to estimate probabilistic constraints in reliability-based topology optimization under random field loads. The combination of Kriging surrogates for the nominal performance landscape with Monte Carlo propagation through the surrogate for uncertainty characterization — avoiding repeated high-fidelity evaluations — is the computational strategy adopted in this study.^{[6][7]}

2.5 Reliability-Based Structural Design

Reliability-based design optimization (RBDO) extends deterministic optimization by constraining the probability of limit-state violation to an acceptable level. Under buckling and compliance constraints, RBDO has been implemented for continuum topology optimization using Kreisselmeier–Steinhauser (KS) constraint aggregation and modified chaos control single-loop approaches. A 2024 study employing surrogate models for reliability-oriented buckling topology optimization under random field loads demonstrated that load uncertainty significantly alters the optimized topology relative to the deterministic solution. Non-probabilistic approaches employing bounded field models provide an alternative when precise probability distributions are unavailable. For manufacturing-induced geometric uncertainty in additive-manufactured metamaterials, however, truncated normal distributions are physically motivated and form the basis of the probabilistic approach adopted here.

Buckling-constrained RBDO remains substantially less developed than compliance and stress-based formulations, confirming the novelty of the present multi-limit-state formulation.^{[5][8][4][6][^1]}

3. Theoretical Framework

3.1 Geometry Parametrization

The baseline specimen is a rectangular coupon of fixed outer dimensions $L \times W \times t$, containing a periodic rectangular array of elliptical through-thickness perforations. Each unit cell is parameterized by five descriptors:

Parameter	Symbol	Physical Meaning
Void fraction	ϕ	Ratio of hole area to unit cell area
Aspect ratio	$\eta = a/b$	Ratio of semi-axes
Orientation	θ	Angle of major axis to loading direction
Pitch ratio	p_x/p_y	Ratio of center-to-center spacings
Ligament ratio	λ	Minimum wall width normalized by cell size

Feasibility constraints enforce $\lambda \geq \lambda_{\min}$ (printability), no hole overlap, and $\phi \in [\phi_{\min}, \phi_{\max}]$.

For functional grading (Scope B), the porosity field follows a polynomial law:

$$\phi(\xi) = c_0 + c_1\xi + c_2\xi^2, \xi = x/L \quad (1)$$

or, alternatively, a discrete three-zone approximation with porosities $\phi_1 \leq \phi_2 \leq \phi_3$ assigned to support, mid-span, and low-demand zones respectively. The grading coefficients $\{c_0, c_1, c_2\}$ or zone values $\{\phi_1, \phi_2, \phi_3\}$ augment the Scope A design vector.

A physics-informed grading variant assigns local porosity via normalized field indicators derived from FE outputs:

$$\psi_k = w_s s_k + w_b b_k + w_m m_k, w_s + w_b + w_m = 1 \quad (2)$$

$$\phi_k = \phi_{\min} + (\phi_{\max} - \phi_{\min})(1 - \psi_k)^\alpha \quad (3)$$

where s_k, b_k, m_k are zone-level normalized static stress severity, buckling mode amplitude, and modal kinetic energy respectively, and α controls grading sharpness. This formulation reduces the grading design space to four parameters $\{w_s, w_b, w_m, \alpha\}$ while remaining physically interpretable and mathematically rich.^[^1]

3.2 Cell-Scale Computational Homogenization

For a representative unit cell Ω_c , the volume-averaged stress and strain tensors are:

$$\bar{\sigma} = \frac{1}{|\Omega_c|} \int_{\Omega_c} \sigma \, d\Omega, \bar{\varepsilon} = \frac{1}{|\Omega_c|} \int_{\Omega_c} \varepsilon \, d\Omega \quad (4)$$

The effective stiffness tensor is defined through:

$$\bar{\sigma} = \mathbb{C}^{\text{eff}} : \bar{\varepsilon} \quad (5)$$

For a 2D orthotropic approximation, four independent components are extracted from three loading configurations applied via kinematically uniform boundary conditions:[^1]

- **Axial x-loading:** $E_x^{\text{eff}} = \bar{\sigma}_{xx} / \bar{\varepsilon}_{xx} = R_x / (A \cdot \Delta / L_c)$
- **Axial y-loading:** $E_y^{\text{eff}} = \bar{\sigma}_{yy} / \bar{\varepsilon}_{yy}$
- **Shear loading:** $G_{xy}^{\text{eff}} = \bar{\tau}_{xy} / \bar{\gamma}_{xy}$
- **Lateral contraction:** $\nu_{xy} = -\bar{\varepsilon}_{yy} / \bar{\varepsilon}_{xx}$ under axial x-loading

An anisotropy index is defined as:

$$A = \max \left(\frac{E_x^{\text{eff}}}{E_y^{\text{eff}}}, \frac{E_y^{\text{eff}}}{E_x^{\text{eff}}} \right) \quad (6)$$

Dimensionless property maps are reported as E_x^{eff}/E_s , E_y^{eff}/E_s , G_{xy}^{eff}/E_s , and ρ^*/ρ_s , enabling density-normalized comparison across the full parameter space.[^1]

3.3 Structure-Scale Performance Indices

All outputs are normalized by a solid (unperforated) baseline under identical boundary conditions and loading:

$$\hat{w} = \frac{w_{\text{max}}}{w_{\text{max,solid}}}, \hat{\lambda} = \frac{\lambda_{cr,1}}{\lambda_{cr,1,\text{solid}}}, \hat{f}_1 = \frac{f_1}{f_{1,\text{solid}}}, \hat{m} = \frac{m}{m_{\text{solid}}} \quad (7)$$

Mass-normalized performance indices suitable for multi-objective optimization are defined as:

$$I_{\text{stiff}} = \frac{1/\hat{w}}{\hat{m}}, I_{\text{buck}} = \frac{\hat{\lambda}}{\hat{m}}, I_{\text{modal}} = \frac{\hat{f}_1}{\hat{m}^{1/2}} \quad (8)$$

The $\hat{m}^{1/2}$ exponent in the modal index reflects the theoretical scaling of natural frequency with (stiffness/mass)^{1/2}.[^1]

3.4 Surrogate Modeling

A Gaussian process (Kriging) surrogate is independently fitted to each scalar performance index:

$$y(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')) \quad (9)$$

where $m(\mathbf{x})$ is the mean function (polynomial trend) and $k(\mathbf{x}, \mathbf{x}')$ is a Matérn-5/2 covariance kernel. The surrogate is trained on 30–40 initial Latin-hypercube samples and iteratively enriched with 10–15 adaptive points selected by expected improvement criteria.^[6]

3.5 Multi-Objective Optimization

The three-objective optimization problem is:

$$\max_{\mathbf{x} \in \mathcal{X}} \{I_{\text{stiff}}(\mathbf{x}), I_{\text{buck}}(\mathbf{x}), I_{\text{modal}}(\mathbf{x})\} \quad (10)$$

subject to:

$$\sigma_{\max}(\mathbf{x}) \leq \sigma_{\text{allow}}, \lambda_{cr,1}(\mathbf{x}) \geq \lambda_{\min}, \lambda(\mathbf{x}) \geq \lambda_{\min, \text{print}}, \phi(\mathbf{x}) \leq \phi_{\max} \quad (11)$$

NSGA-II is employed as the multi-objective evolutionary optimizer acting on the surrogate surface, with population size of 200 and 500 generations.^[1]

3.6 Uncertainty Quantification and Robust Optimization

Uncertain geometric parameters are modeled as independent truncated normal random variables:

$$\mathbf{X} = \{a, b, \theta, \lambda, E_s\} \quad (12)$$

Three performance limit states are defined:

$$g_1(\mathbf{X}) = \sigma_{\text{allow}} - \sigma_{\max}(\mathbf{X}) \quad (13)$$

$$g_2(\mathbf{X}) = \lambda_{cr,1}(\mathbf{X}) - \lambda_{\min} \quad (14)$$

$$g_3(\mathbf{X}) = f_1(\mathbf{X}) - f_{\text{target}} \quad (15)$$

The system-level failure event is:

$$g(\mathbf{X}) = \min \{g_1(\mathbf{X}), g_2(\mathbf{X}), g_3(\mathbf{X})\} < 0 \quad (16)$$

Failure probability and reliability index are computed as:

$$P_f = \Pr [g(\mathbf{X}) < 0], \beta = -\Phi^{-1}(P_f) \quad (17)$$

where Φ^{-1} is the inverse standard normal CDF. Robust Pareto fronts are obtained by optimizing the robust performance index:[⁴][1]

$$J_{\text{rob}} = \mu(J) - \kappa \sigma(J) \quad (18)$$

where κ is a risk-aversion coefficient and $\mu(J)$, $\sigma(J)$ are the mean and standard deviation of a performance index under the uncertainty distribution.

Global variance-based sensitivity analysis employs Sobol indices:[⁶]

$$S_i = \frac{\text{Var}_{X_i}(\mathbb{E}[Y | X_i])}{\text{Var}(Y)} \quad (19)$$

with first-order indices S_i and total-effect indices S_{T_i} estimated via the Saltelli–Sobol estimator from surrogate evaluations.

For studies seeking additional mathematical depth, ligament-width spatial variability is modeled as a Karhunen–Loève random field:

$$t(x, \omega) = \bar{t}(x) + \sum_{n=1}^M \sqrt{\lambda_n} \xi_n(\omega) \phi_n(x) \quad (20)$$

where λ_n and $\phi_n(x)$ are eigenvalues and eigenfunctions of the covariance kernel, and $\xi_n(\omega)$ are uncorrelated unit-variance random variables.[⁶]

4. Methodology

4.1 Phase 0: Literature Mapping and Benchmarking

A structured literature matrix of 30–40 papers is compiled with columns for topology type, material system, physics studied (static / buckling / modal), uncertainty treatment, experimental validation, optimization method, key gap, and reusable results. Anchor clusters cover computational metamaterial design, perforated plate lattices, multiscale homogenization, and reliability-based optimization.[¹⁴][7][¹³][2][³][9][⁴][6]

4.2 Phase 1: Model Verification and Benchmarking

Credibility of the FE workflow is established in three sequential steps before any parametric campaign.[¹]

Step 1.1 — Solid baseline. An unperforated coupon is analyzed under the same boundary conditions as the study domain. Static deflection is compared against Euler–Bernoulli beam theory or Kirchhoff plate theory; the first buckling multiplier is compared against classical Euler or plate-buckling analytical formulae; and the first natural frequency is compared against beam or plate vibration theory.

Step 1.2 — Simple perforated baseline. A single circular-hole specimen is introduced and a systematic mesh convergence study performed. Compliance is monitored as a convergence metric for static studies; von Mises peak stress is tracked at a fixed spatial location rather than the global maximum (which migrates with mesh refinement). For buckling, manual mesh convergence is required because adaptive refinement is not available in the Fusion 360 buckling solver. Displacement typically converges faster than stress in FE analyses, and this hierarchy is respected in reporting.^[1]

Step 1.3 — Physics-specific cautions. Eigenvalue buckling implemented in Fusion 360 is based on fully linear elastic assumptions; a high buckling multiplier alone does not constitute a safety guarantee, and static and buckling results must be interpreted together. For modal studies, unconstrained models introduce rigid-body modes; design standards typically require 85–90% cumulative mass participation as a stopping criterion for mode count. Symmetry models are explicitly avoided for both modal and buckling runs, as they can suppress physically relevant asymmetric modes.^[1]

4.3 Phase 2: Parametric Geometry Family Construction

The verified baseline is extended to the five-parameter elliptical perforation family described in Section 3.1. Constraints — minimum ligament width, no overlapping holes, and minimum printable feature size — are enforced at the CAD generation stage. Geometry is constructed parametrically in Fusion 360, enabling automated regeneration for each design vector sample.

4.4 Phase 3: Cell-Scale Property Extraction

A representative unit cell, or a 3×3 supercell to reduce boundary-condition bias, is subjected to the three loading configurations described in Section 3.2 via prescribed displacement boundary conditions. Reaction forces and applied displacements yield the effective stiffness components through the formulae in Eq. (4)–(5). The anisotropy index *A* (Eq. 6) is computed for each cell geometry and stored alongside structural-scale outputs for cross-scale correlation analysis.

4.5 Phase 4: Structure-Scale Simulation Campaign

Every sample design is analyzed under three FE study types:

Study Type	Primary Outputs
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Static stress	Max deflection $w_{\max}$, compliance $C = \mathbf{F}^T \mathbf{u}$, peak von Mises stress, strain energy
Structural eigenvalue buckling	First buckling multiplier $\lambda_{cr,1}$, first 3–5 mode shapes
Free vibration (modal)	First 6–12 natural frequencies, mode shapes, directional mass participation factors

All outputs are normalized by the solid baseline values per Eq. (7) and mass-normalized performance indices computed per Eq. (8).

4.6 Phase 5: DOE, Surrogate Fitting, and Pareto Optimization

A Latin hypercube sample of 30–40 points covering the feasible five-parameter space is generated using a maximin criterion. Following initial FE evaluation, 10–15 adaptive enrichment points are selected by expected improvement. Gaussian process surrogates (Eq. 9) are independently fitted to I_{stiff} , I_{buck} , and I_{modal} , and leave-one-out cross-validation is used to assess fit quality. NSGA-II (Eq. 10–11) is executed on the surrogate surface with a population of 200 over 500 generations to identify the Pareto front in the three-objective space. Top Pareto designs are verified by direct FE re-analysis to validate surrogate fidelity.

4.7 Phase 6: Functional Grading Extension

The best uniform Pareto family is extended by introducing the grading parameters of Eq. (1)–(3). Graded specimens are generated by assigning spatially varying porosity fields and re-meshing. Both polynomial grading (Eq. 1) and physics-informed grading (Eq. 2–3) laws are evaluated. The augmented Pareto front — uniform and graded designs together — is used to test Hypothesis H1 and H3.

4.8 Phase 7: Robust Optimization and Sensitivity Analysis

Morris elementary effects screening is applied first to identify non-influential uncertain parameters. The reduced uncertainty set is propagated through the trained Kriging surrogate via Monte Carlo sampling (10,000 samples per design) to estimate $\mu(J)$, $\sigma(J)$, P_f , and β (Eq. 16–17). Robust Pareto fronts are generated by NSGA-II optimizing J_{rob} (Eq. 18). Sobol first-order and total-effect indices (Eq. 19) are estimated for the Pareto-optimal nominal and robust designs, yielding a quantitative sensitivity ranking of uncertain parameters. A Karhunen–Loève spatial random field (Eq. 20) is optionally applied to ligament-width variability as an advanced extension.^{[7][6]}

4.9 Phase 8: Optional Low-Cost Experimental Validation

Three specimens — the solid baseline, the best nominal-optimized design, and the best robust-optimized design — may be 3D-printed and subjected to:

- Mass measurement (verify relative density);
- Static three-point bending under known loads (verify deflection trend);
- Tap-test frequency measurement via smartphone FFT app (verify first modal frequency rank-order).

Validation claims are intentionally bounded to qualitative trend and rank-order agreement rather than quantitative predictive fidelity, consistent with the nature of the test instrumentation.^[1]

5. Hypotheses

Three falsifiable hypotheses structure the study and define the decision criteria for Scope B:^[1]

H1. A modestly anisotropic perforation family with longitudinal functional grading will outperform the best uniform perforation design at equal mass in the combined static–buckling–modal objective space.

H2. The nominal deterministic optimum will be more extreme in geometric parameters and more sensitive to manufacturing variation, while the robust optimum will sacrifice some nominal performance but substantially reduce output variance and failure probability.

H3. A low-dimensional grading law — either the three-zone or quadratic polynomial formulation — can capture most of the performance gain achievable by a much higher-dimensional hole-by-hole design space.

These hypotheses are non-trivially falsifiable: a negative result on H1 still yields a publishable finding demonstrating the limits of grading benefit in this geometry class; a negative result on H2 demonstrates robustness at near-zero nominal cost; and a failure of H3 motivates higher-order grading laws.^[1]

6. Expected Results and Contributions

6.1 Scope A Anticipated Outcomes

The design-of-experiments campaign and surrogate-Pareto analysis are expected to yield:

- **Performance maps** showing how I_{stiff} , I_{buck} , and I_{modal} vary across the five-parameter space, including nonlinear interaction effects between void fraction ϕ and orientation θ .
- **Pareto fronts** in the three-objective space, quantifying the inherent trade-offs between stiffness, buckling resistance, and vibration performance at equal mass.

- **Sensitivity rankings** identifying ϕ and θ as dominant drivers of the stiffness–buckling trade-off (anticipated based on existing literature), with η controlling the degree of performance anisotropy.^{[3][9]}
- A **"best uniform family"** — a cluster of Pareto designs sharing geometric heuristics — suitable as design guidelines.

6.2 Scope B Anticipated Outcomes

The multiscale and robust extensions are expected to produce:

- **Effective stiffness tensor maps** E_x^{eff}/E_s , E_y^{eff}/E_s , G_{xy}^{eff}/E_s as functions of ϕ , η , and θ , providing structure–property relationships beyond single-output FE studies.
- **Anisotropy–performance correlation:** a positive correlation between anisotropy index A and buckling multiplier $\lambda_{cr,1}$ is anticipated for load-aligned perforation orientations, consistent with the theoretical role of anisotropy in delaying buckling.^[^3]
- **Grading benefit quantification:** the polynomial or physics-informed grading law is expected to yield measurable improvement in I_{buck} and I_{modal} at equal or lower mass relative to the best uniform design, consistent with existing literature on buckling-driven hole placement.^{[15][3]}
- **Robust vs. nominal trade-off curve:** the cost in mean performance of achieving a target reliability index $\beta \geq \beta_{\text{min}}$ is expected to be modest (estimated 5–15% degradation in J_{rob}), consistent with analogous findings in continuum RBDO.^{[4][6]}
- **Dominant sensitivity finding:** ligament ratio λ and hole semi-axis a are anticipated to dominate buckling-related Sobol total-effect indices, while orientation θ is expected to dominate stiffness-related variance, providing actionable manufacturing tolerance recommendations.^[^1]

6.3 Novel Contributions

The aggregate novel contributions of this work are:

1. A **verified, reproducible Fusion 360-based FEM workflow** for three-physics (static, buckling, modal) evaluation of perforated plate metamaterials, with documented mesh convergence and physics-specific cautions.
2. **Cell-scale effective stiffness tensor maps** and an anisotropy index correlating hole geometry to orthotropic mechanical properties within an accessible CAD-driven FEM environment.
3. **Mass-normalized Pareto fronts** for simultaneous stiffness, buckling, and modal performance — a characterization dimension absent from prior perforated-plate literature.^{[3][1]}

4. A **low-dimensional graded perforation law** — polynomial and physics-informed variants — demonstrated to outperform or match uniform designs at equal mass.^[1]
5. **Robustness maps and Pareto fronts under geometric uncertainty**, quantifying the nominal–robust trade-off and failure probability for multi-limit-state performance.^{[4][6]}
6. **Sobol sensitivity rankings** identifying which manufacturing tolerances most critically govern structural reliability.^{[7][6]}

7. Discussion

7.1 Framework as the Contribution

A critical conceptual point is that the publishable novelty of this work is not a new hole shape or a single winning geometry. The contribution is the **framework**: a disciplined, reproducible, robustness-aware design procedure that maps geometric parameters to multi-physics performance under uncertainty. This framing is consistent with the emerging emphasis in the mechanical metamaterials community on inverse design frameworks and data-driven methods, and directly addresses the identified gap in practical homogenization-based operational workflows.^{[2][1]}

7.2 Scope and Limitations

Several limitations bound the scope of conclusions. The FE analyses assume linear elastic isotropic base material properties, which is appropriate for the geometry-driven performance claims but precludes extension to post-yield or nonlinear constitutive behavior without supplementary studies. Eigenvalue buckling is a conservative linear elastic approximation; post-buckling reserve capacity and geometric imperfection sensitivity would require nonlinear (Riks) analysis. The homogenization is first-order and assumes scale separation; for very large holes relative to the specimen, higher-order or generalized continuum theories would be needed. Experimental validation, if included, is deliberately limited to trend and rank-order agreement rather than quantitative fidelity, given the instrumentation constraints.^{[13][9][1]}

7.3 Scalability and Transferability

The framework is intentionally solver-agnostic at the mathematical level; although Fusion 360 is used as the FE engine in this work for its accessibility and integrated parametric CAD environment, the homogenization procedure, DOE strategy, surrogate fitting, and UQ propagation workflow can be directly transferred to any FE platform (Abaqus, ANSYS, COMSOL) for higher-fidelity extensions. The physics-informed grading law (Eq. 2–

3) is portable to non-elliptical perforation families and to three-dimensional plate lattices, including SC, BCC, and FCC topologies such as those studied in the perforated plate lattice literature.^[3]

7.4 Roadmap for Future Work

The most natural extensions of this study are:

- **Nonlinear post-buckling analysis** to quantify post-critical load-carrying capacity for the optimal graded designs.
- **3D anisotropic homogenization** extending the 2D orthotropic tensor extraction to full three-dimensional plate lattice cells.
- **AM process-aware uncertainty modeling** linking specific LPBF process parameters (laser power, scan speed, layer thickness) to dimensional tolerances via a process–structure linkage model.
- **Experimental characterization** of the Pareto-optimal designs under standardized mechanical testing protocols to provide quantitative validation.
- **Phononic and acoustic metamaterial extension** exploring how the stiffness anisotropy maps produced in this work translate to bandgap engineering.

8. Conclusion

This paper presents a reliability-aware multiscale framework for the design of functionally graded perforated plate metamaterials evaluated simultaneously under static, buckling, and modal constraints. The framework integrates five key components — parametric geometry with elliptical perforations, cell-scale computational homogenization, structure-scale Fusion 360 FE analysis, Gaussian process surrogate modeling with NSGA-II multi-objective optimization, and surrogate-based uncertainty quantification with Sobol sensitivity analysis — into a layered, modular workflow. The approach is designed to be resilient: each layer constitutes an independent publishable contribution, and no single failure point can invalidate the overall study.^[1]

Anticipated results will demonstrate that functional grading of the porosity field enables performance gains in the combined static–buckling–modal trade space at equal mass, that robust optimization produces designs with substantially lower output variance and failure probability at modest nominal cost, and that ligament width and hole dimensions dominate manufacturing-induced performance uncertainty. The Sobol sensitivity rankings translate directly into actionable, quantified manufacturing tolerance recommendations — a form of design guidance absent from existing perforated plate metamaterial literature.^{[6][3][1]}

The project sits at the intersection of mechanical metamaterials, multiscale mechanics, lightweight structures, uncertainty quantification, and additive-manufacturing-aware design — a combination identified in recent reviews as an important open research frontier.^{[12][2][^1]}

Acknowledgements

[To be completed: Funding sources, computational resource acknowledgements, institutional support.]

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Additional references to be added from Phase 0 literature matrix, particularly covering:

- *Computational metamaterial design reviews (Nature Computational Science, 2024)*
- *Lattice mechanics and static/dynamic behavior reviews (Progress in Aerospace Sciences)*
- *NSGA-II original algorithm paper (Deb et al., 2002)*
- *Gaussian process regression (Rasmussen & Williams, 2006)*
- *Sobol sensitivity analysis (Saltelli et al., 2010)*
- *Karhunen–Loève expansion for random fields (Ghanem & Spanos, 1991)*
- *Autodesk Fusion 360 Simulation documentation*

References

1. FEM_project.pdf - page-1
Recommended project Title: Reliability-aware multiscale design of functionally graded per...
2. Mechanical metamaterials and beyond - Mechanical metamaterials enable the creation of structural materials with unprecedented mechanical p...
3. [PDF] Metallic perforated plate lattices with superior buckling strength - Inspired by this idea, we propose a method to design the optimized micro hole locations for perforat...
4. Reliability-based topology optimization of continuum structure under ... - A reliability-based topology optimization (RBTO) model under buckling and compliance constraints is ...
5. Manufacturability of functionally graded porous β -Ti21S auxetic ... - The as-designed and the as-printed strut thickness values were used to calculate the homogenized pro...
6. Surrogate model-based method for reliability-oriented buckling ... - This paper develops a surrogate model-based method for reliability-oriented buckling topology optimi...
7. Surrogate model-based method for reliability-oriented buckling topology optimization under random field load uncertainty - Structures that are subjected to compressive loads may experience not only local fractures but also ...

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14. [\[PDF\] Mechanics and Computational Homogenization of Effective Material ...](#) - This study delves into various methods and theories for modeling and analyzing functionally graded m...
15. [Buckling analysis of perforated stiffened composite plates with ...](#) - This study focuses on the buckling behaviour of stiffened perforated laminated plates with interfaci...
16. [Reliability-based topology optimization of imperfect structures ...](#) - In this paper, a novel optimization technique is implemented to explore the effects of considering u...